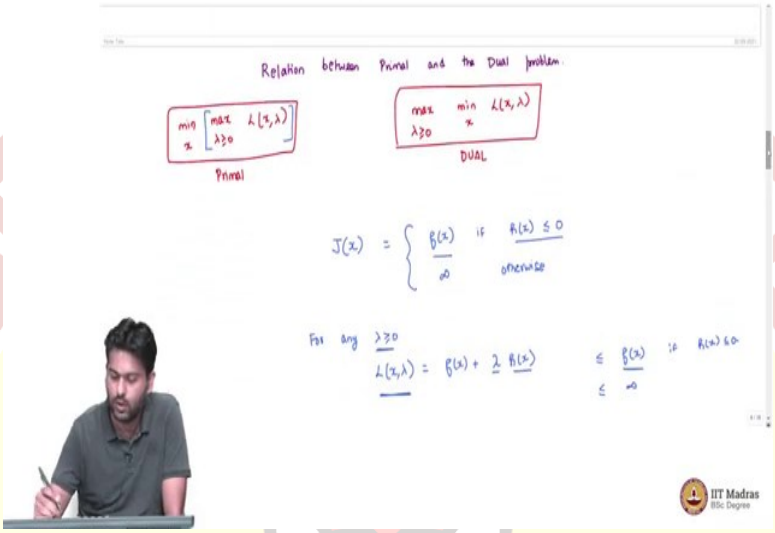


IIT Madras

ONLINE DEGREE

MLF
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Relation between Primal and Dual Problem, KKT Conditions

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So, what we want to do now is look at the relation between primal and the dual problem. What was the primal problem? The primal problem was max over n over x, max over λ greater than 0, L of x , λ, this was the primal problem. The dual problem, this is primal. The dual is max over $\lambda \geq 0$ min over x, L of x , λ. This is the dual problem.

And we want to understand how these two things are related. Now, to understand this, let us look at the one more definition of a different function. Let us call this function J. So, this is a function again, this is a function of x. So, what is it going to, so how are we going to define this function? Well, we are simply going to say this function is going to be f of x if h of x is $\leq$ 0, and infinity otherwise, this is very closely related to the earlier function that we looked at.

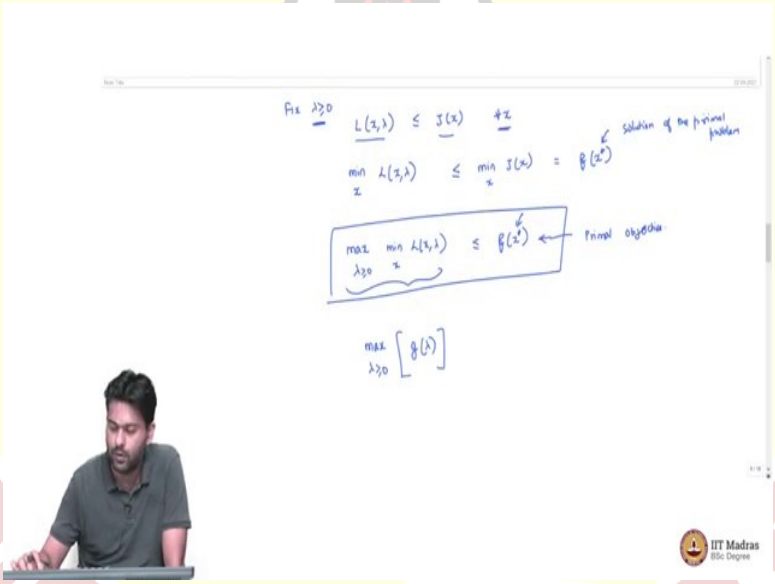
Just that we are now saying that instead of saying f of x + λ h of x, which is the Lagrangian function, we are just directly trying to look at a function J(x), where J(x) equals f of x if h of x is $\leq$ 0 or infinity otherwise. Of course, you can think of this as maximizing over λ, our Lagrangian for a given x , λ also. So anyway, so this is, let us say we think of this as a separate function.

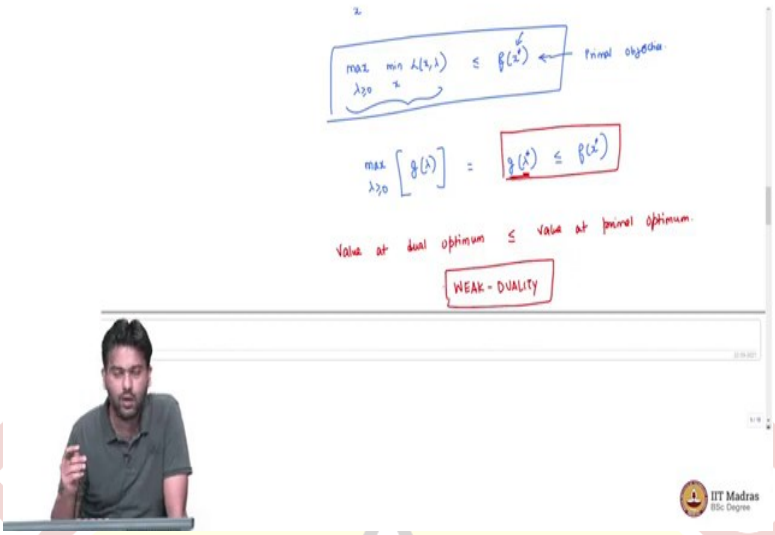
And now, if you look at for any λ, which is ≥ 0, so let us say fix a λ i, and they ask, what is L of x , λ Lagrangian, that is going to be simply f of x + λ h of x. Now, if h of x is ≤ 0, then this whole value is going to be f of x minus something. So, because we have fixed λ is ≥ 0.

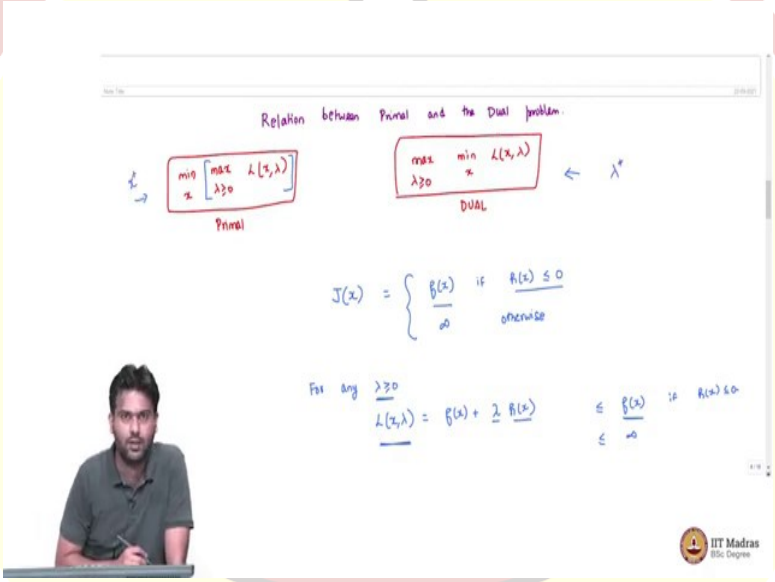
Our Lagrangian function for a given λ is going to take a value which is f of x + a positive or a non-negative value times a value which is non positive. So, this is going to be $\leq$ f of x, which means that this is going to be if this is going to be $\leq$ f of x, if h of x is ≤ 0.

So, on the other hand, if, if h of x is ≥ 0, well, this, this, this value is going to be f of x + some positive or a non-negative value, which is still going to be $\leq$ infinity. Well, why am I putting it this way? The reason is, now, you can you can think of this L of x , λ, as you know some kind of a linear function which is lower bounding the function J(x).

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So, in other words, what I want to convey here is that our L of x , λ for, so, we have fixed $\lambda \geq 0$, and L of x , $\lambda\lambda$ is $\leq$ J(x) for all x, no matter where x is. So, the only two interesting possibilities are whether x is such that h of x is ≤ 0.

It satisfies the constraint or does not satisfy the constraint. If it satisfies the constraint, it is this, the Lagrangian function is still $\leq$ f of x. If it does not satisfy the constraint, it is still $\leq$ infinity, which means it is $\leq$ J(x). So, this is always going to be $\leq$ J(x).

Which means we can say that if I look at the minimum over x, so this is for all x, which means in particular it has to hold for the minimum value; the minimum over x L of x , λ has to be $\leq$ minimum over x, J(x) this has to happen. Now, now let us think about what is the minimum

over x J(x). Remember J(x) is defined such that if the constraint is satisfied this is f of x otherwise it is infinity.

Which means that my minimum is simply going to be f of x star, it is where x star is like the optimal solution of the original problem that they wanted to solve. Now, remember this held for any λ, I picked an arbitrary λ and I said but my Lagrangian function is lower bounding the J(x) function. So, now which means that I can, because it holds for any λ it also has to hold necessarily for the λ that maximizes this function.

So, which means that max over λ are ≥ 0 min over x L of x , λ is also $\leq$ f of x star. But what does this tell us this tells us something interesting. So, this f of x star where x, what was x star? x star is kind of the solution of the primal problem. So, remember, this is the solution of the primal problem. What is the primal problem just to remind you, the primal problem is the original problem that we wanted to… is equivalent to the original problem that we wanted to solve.

Which means that if I solve this problem and find an x star, and if we look at the functions value that will be the smallest possible value in within the constraint set. Now, the interesting part is that we have shown that the maximization over $\lambda \geq 0$ minimization of x, which means that we have swapped these two guys is still $\leq$ f of x star.

So, this is the primal objective so and what is this? So, this is simply maximize over $\lambda \geq 0$ by our definition, this is g of λ, where we just defined g of λ as minimization over x, L of x , λ. Now, eventually, what are we essentially saying. So, which means that if I had solved the dual problem, which is here and I had gotten λ star, if I solved the primal problem, I had gotten x star.

Now, what we are saying is that this is simply g of λ star and this is $\leq$ f of x star in words. All we are saying is that the value at dual optimum is $\leq$ the value at the primal optimum. This is very, very interesting. So, what would what did we do so, just to recap, we started with the constrained optimization problem.

We converted it into a problem which was of the form min max and we said that these two are equivalent and that we call the primal problem and we swap the min and the max, we call the dual problem. Now, if you solve the dual problem, you will get some λ star and if you look at the functions value at λ star, well that functions value is in fact $\leq$ the primal functions value,

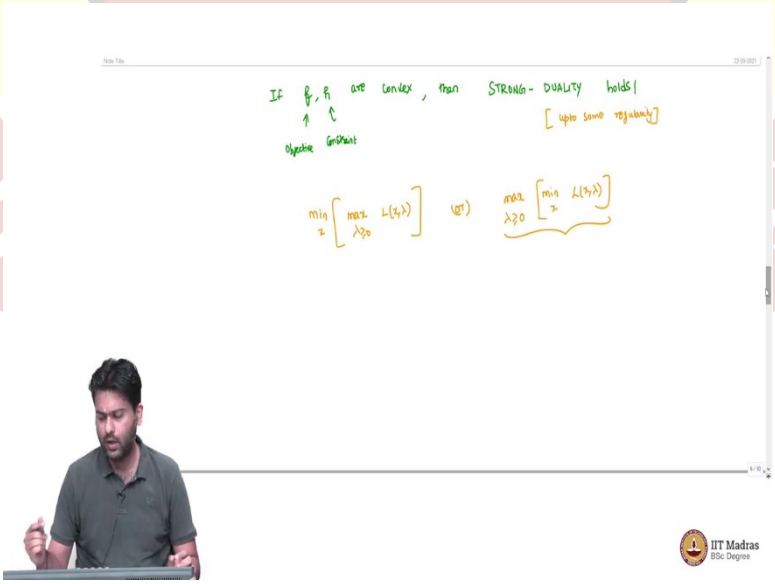
which is optimal value, which is f of x star, we know that is f of x star. So, what we are saying is that well the solution…

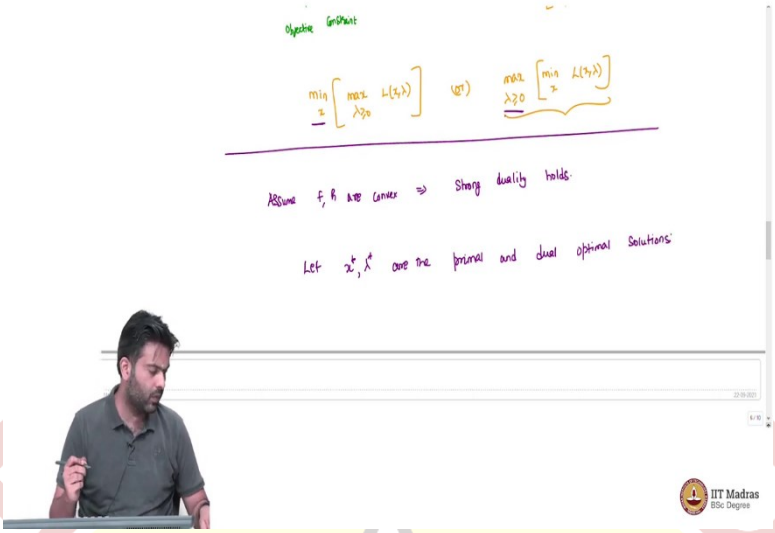
So, if you solve the dual problem and look at the functions value of the dual functions value, it may not be equal to the primal functions value, which means that it does not mean that you have solved the primal problem. We know the primal is equivalent to the constraint optimization problem, we set out to solve. Now, if you solve the dual, all we are guaranteed is that g of λ star the duals value is $\leq$ the primal's value.

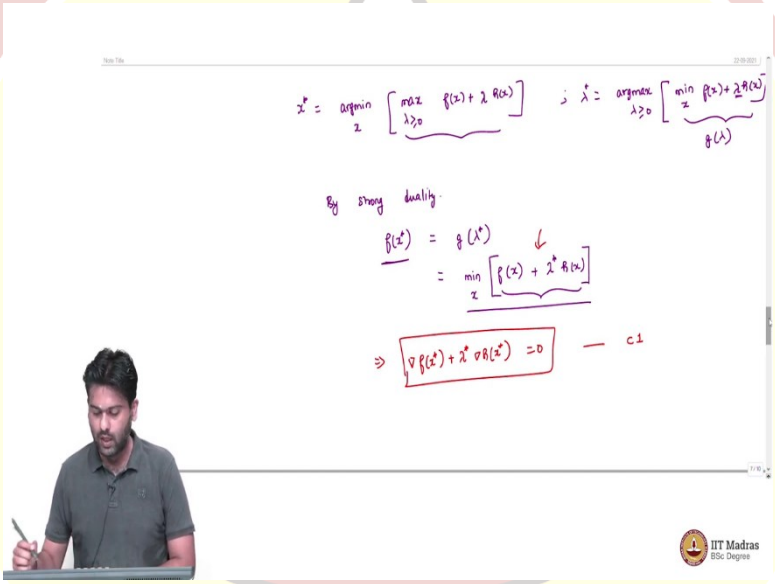
So, this condition is what is called as a weak duality. So, which means that you are dual problem which is easy to solve gives a quick I mean it easy lower bound for the primal optimal solution. So, which is which is useful in a lot of cases, it turns out. So, we will not discuss those cases in detail now.

But all I am trying to say is that in general without making any further assumptions about the nature of the functions f or the constraint h or anything of that sort we are saying that we can take any problem, convert it into a concave maximization problem, where the constraints are $\lambda \geq 0$, called the dual problem, solve the dual problem, the objective value of the dual solution is $\leq$ the primal optimal solution which is which is the relation between the dual and the primal?

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Now, the very interesting part again there is without making a very rigorous statement, let me give you the most interesting statement, which is which will be relevant to the machine learning part later. Now, so if the functions that we are trying to look at, so, the objective function, so, this is the objective function and this is the constraint. If these functions are convex, then something very interesting happens.

So, you do not just have weak duality, you have what is called a strong duality. Strong duality holds. This is a very, very powerful theorem. So, we will not prove this theorem. And let me let me just say that this holds up to some regularity conditions, some regularity, which is not the focus of this broad overview. Your x, the optimal points should satisfy some regularity conditions.

But the main structure, the assumption that leads to this is the convexity of f and h. So, let us focus on that. So, if our function that we are trying to optimize is a convex function, and the constraint that we have is also a convex constraint, then what this is saying is that you can either solve the primal problem or solve the dual problem.

So, the objective value that you will get by either solving the primal problem and the dual problem would exactly be the same. That is what strong duality says. So, in other words, you can either solve min over x, max over λ greater than 0 L of x , λ or you can solve max over $\lambda \geq 0$ min over x L of x.

And in a lot of cases, this may be easier to solve. So, I am not saying it is always easier to solve, there may be cases and in machine learning, we will see why not will later see when you are discussing some algorithms why this is an easier problem to solve. And I will talk a bit about the algorithms that are well such a duality view for helps a lot after we finished the discussion about constraint optimization. So, all we are saying is that which is very good. So, if f and h are convex, then strong duality holds, which means that we can either solve the primal problem or solve the dual problem.

But in general, if you solve the dual problem, the solutions value or the dual optimal is going to be $\leq$ the solutions value at the primal optimal. In special case of convexity these two things happen to be similar same. So, this is the first main result and we will now in the next few minutes, see, what are the implications of this result. So, what can we understand from result?

Remember, initially, we set out to understand well, if I have an unconstrained optimization problem, then if I find an x star, where the gradient is 0, then I am done. So, if f is convex, for example, then I know this is a global minimum. Otherwise, I know this is like local optima. Now, here, we want to understand something similar for the constraint optimization problem. That is where we started off with.

And all we are saying so far is that well, we are nowhere at least at this point. We have not really understood what are the conditions that a point should satisfy such that it is necessary or sufficient to be a local optima or global optima, we are not talking about that at all yet. All we are saying so far is that if we want to solve the original constraint optimization problem you can either solve the primal version of it or the dual version of it.

If f and h are convex then these two doing these two things are the same, of course up to some regularity conditions on the on the function on the optimality of the function itself. So, this is where we are so, what we want to do next is that we want to kind of convert this understanding into a set of conditions that the optimal solution will satisfy.

So, that is that is what we want to focus on next. For that, let us let us do the following. Let us assume so, so, let us assume f and h are convex. So, which means that assume f and h are convex so, which means strong duality holds, there are some regularity conditions again I keep mentioning that, but let us forget that for the moment.

So, assume strong duality holds. Now, we have the primal problem we have the dual problem the primal is min over x the duel is a max over λ. L t us say x star , λ star are the primal and dual optimal solutions. So, what do I mean by that? So, just to be absolutely sure because we will use this I am saying x star is simply the arg min over x max over $\lambda \geq 0$, f of x + λ h of x and λ star is arg max over λ greater than 0 mean over x f of x + λ x.

Let us say these are the optimal solutions. So, now, what we are trying to understand is if x star and λ star or optimal solutions any optimal solution for this problem then what conditions should they satisfy. So, what are the necessary conditions for x star and λ star so, to be optimal?

Remember we are assuming the strong duality holds in other words in this case we are assuming f and h are convex strong duality holds and so, we will not understand what can we say about x star and λ star. Of course, we know x star is the argument of this λ star is the argument of the dual problem, but can we say something about the properties that this pair should satisfy?

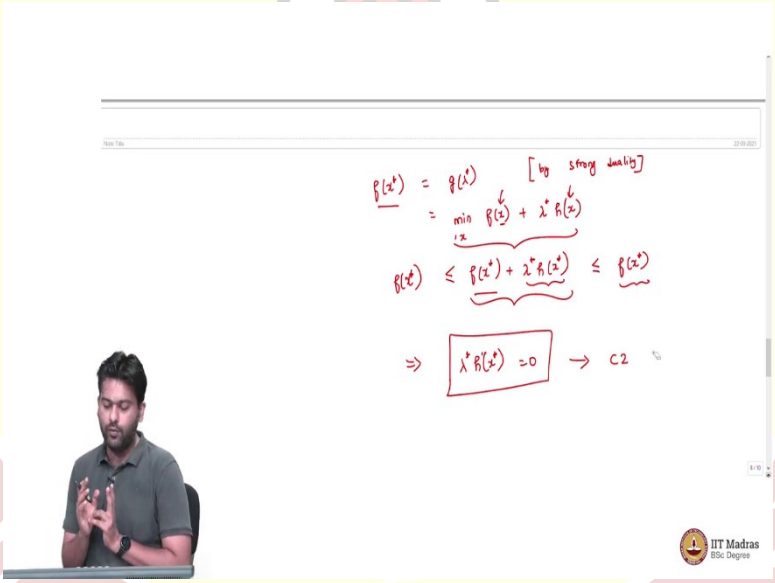
For that let us look at the following. So, now if we look at so, because strong duality holds so, by strong duality; now, my f of x star. So, of course, if I evaluate the functions value at x star that is going to be f of x star. So, that is equal to like g of λ star. So, that is what strong duality kind of tells us. Now, well what is g of λ star? g of λ star is min over x f of x + λ star h of x.

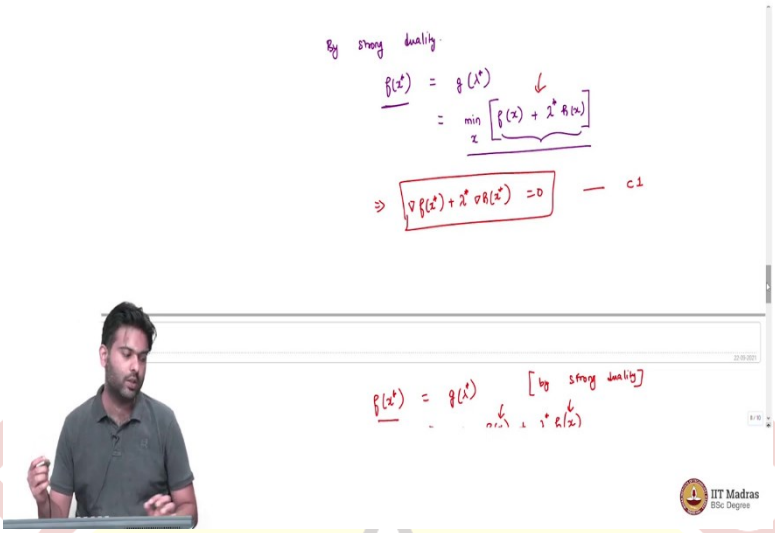
So, so, this is remember this is the g of λ function. Which means that if I plug in λ star into this that just means that this λ I am replacing it by λ star. So, now, what does this tell us? This tells us two important things by strong duality I know f of x star is same as min over x f of x + λ star h of x which means that the functions value at x star is same as the functions value that I would get if I minimize this function.

Well, what does that mean? So, that means that where what should x star be. So, this is a demand for an unconstrained optimization problem. This means the following should then hold. So, if I look at the gradient of this function at x star that gradient should be 0. Because this is an unconstrained optimization problem and because f and h convex our I mean we are looking at the minimizer.

So, which means that the minimizers gradient will be 0. So, which means then what we are saying is that this implies the first important condition for our x star should satisfy is that the gradient of this function which is like the L of x star + λ star they del of h star h of x star equals 0. So, this is let us say C1 condition number one. So, this this this is simply because of strong duality. So, any x star λ star pair that satisfies the optimality, so, necessary has to satisfy these two which is good. Now, something else also is true.

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So, we look at the same exact argument, but then some other conclusion will also be true. So, we say if f of x star equals J(x) star. Now, we just said J(x) star is min over x f of x + λ star g of x. So, which means that this is the minimum over f of x + λ star g of x over all possible x. So, the minimum over all possible x, which means that I can plug in any x here any x dash here.

And f of x dash + λ star g of x dash will be n will be $\geq$ this value. So, now let us say substitute this x as x star itself which means that this has to be $\leq$ because this is minimum over all x in particular for x star also. So, this is $\leq$ f of x star + λ star g of x star.

You see why that happens because any x I can plug in any x and then the minimum is going to be less than that right. So, this is min over all x. Now, this is a surprising fact then. So, what we are saying is that the left-hand side by strong duality. So, this is a strong duality is f of x star the right-hand side seems to be this is upper bounded by f of x star + λ star sorry this should be h, h sorry about that is h did I use g somewhere else as well. λ star h of x star. Well now, what can we, what does this essentially mean that?

So, we are saying f of x star is $\leq$ f of x star + λ star h of x star. But by we also know that λ star has to be ≥ 0; h of x star has to be ≤ 0. So, because at x star h of x star satisfies the constraint, so, h of x star will be ≤ 0 and λ star will be ≥ 0, which means that this whole thing is going to be ≤ 0, which means that this I can upper bound by f of x star.

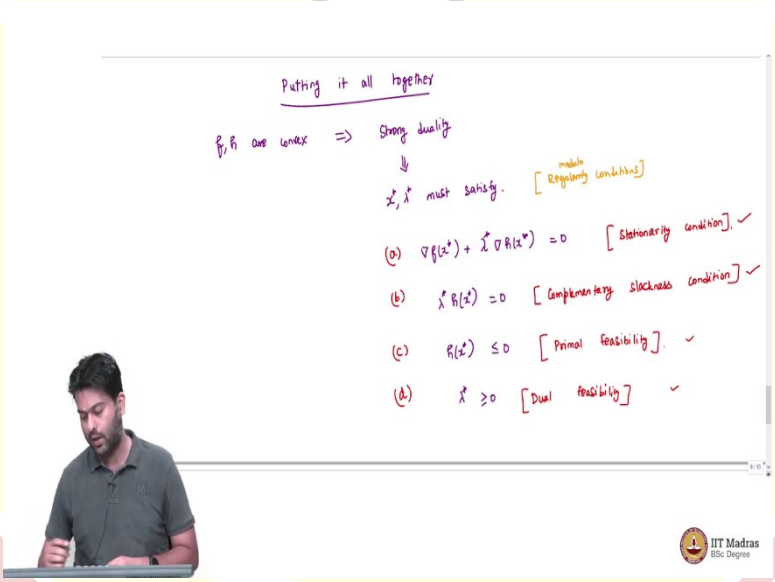
So, this is f of x star + something and that something is ≤ 0. So, the whole thing is upper bounded by f of x star. Now, we started off with f of x star. And we are saying here is a sum of

two terms. And we are upper bounding also by f of x star, which means that the only way this can happen is when all these three things are exactly the same.

So, otherwise this cannot happen, which in in turn means that the only way this can happen is if the second term is 0, which is λ star into h of x star equals 0. Now, here is another condition. This is a very important condition. So, this is C2 this, we will come back to the name a bit later, but yeah, so, this is condition 2. So, we have kind of derive 2 conditions now, on what my x star , λ star pair should satisfy.

If, if f and h are convex. So, any optimal solution necessarily should satisfy these 2 conditions. Now, let me put them all together for you.

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Now, putting all together all what are we saying. So, we are saying the following let us say f and h are convex. Now, this implies strong duality. Now, this implies let us say x star , λ star must satisfy the following condition one which I am rewriting again. So, λ f of x star + λ, del f of x star + λ delta h of x star λ star h of x star is 0 and this is often called as the stationary condition. This is condition one.

Condition two is, well, we saw h of x star into λ star has to be 0 and this is a very important condition and this is called as the complimentary slackness condition. It is called complimentary slackness because I either λ star is 0 or h of x star is 0 or both are 0. So, it cannot be the case in particular that if at optimal h of x star is less than 0, strictly less than 0 is just strictly within the constraint set then λ star cannot be strictly greater than 0.

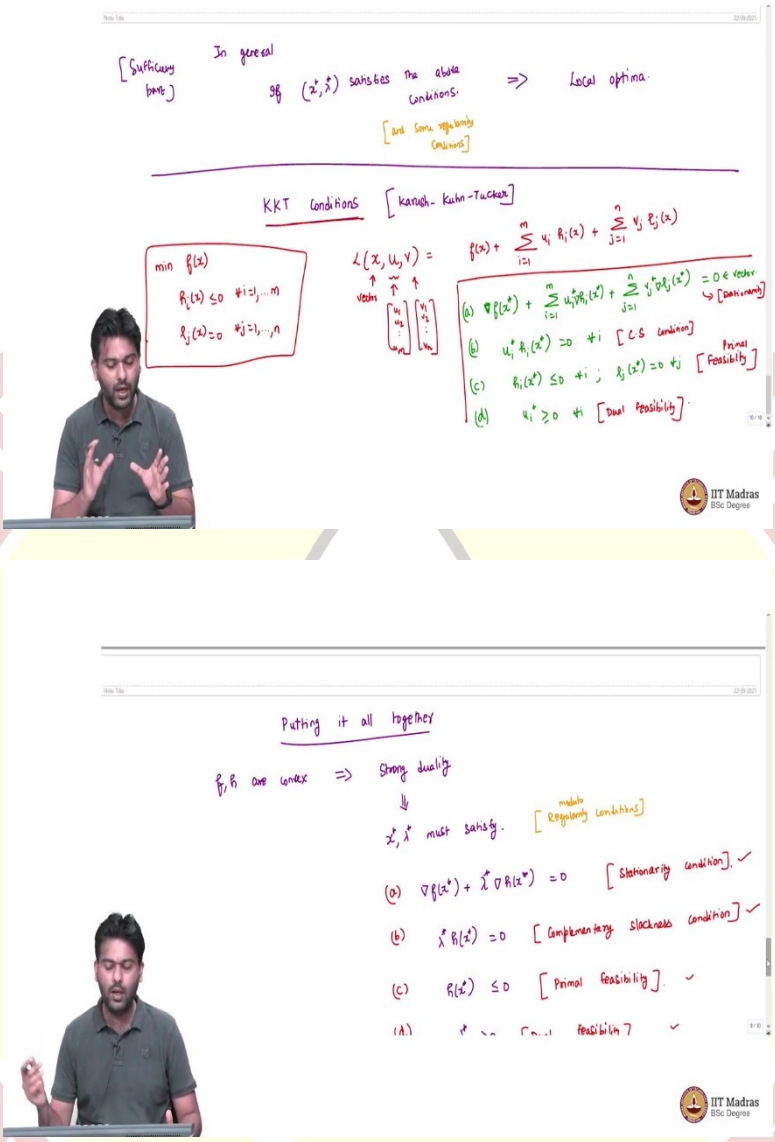
So, then λ star strictly has to be 0. So, that is what this is essentially telling us. It is the product of these two has to be 0 and that is why it is called a complimentary slackness condition. Third point well these are the other things are straightforward, which we already know h of x star has to be ≤ 0, well, that is a like feasibility check. So, your optimal problem point should necessarily also be feasible.

So, this is what is called as a primal feasibility condition. The fourth condition is of course, there is a dual feasibility, what is the dual feasibility? What should λ star satisfy? It should be ≥ 0. So, this is the dual feasibility these are something that we already knew, no, because, well, these are simple. Any feasible point should not necessarily satisfy this.

What is interesting is that now we are also saying that if you have an optimal point, then it necessarily has to satisfy the stationarity condition and also the complimentary slackness condition. Now, of course, these are necessary conditions. So, if f and h are convex, then the optimal point should necessarily satisfy these things. So, of course, again, let me say that the x star and λ star should there should be some regularity conditions.

So, modulo, which are I did not want to get into the rigorous part of it, but that is not the most important part. The most important part are these four conditions are satisfied for any optimality x star and λ star.

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Now, the question one might ask is that well, this is a necessary condition. So if f and h are convex, then the optimal solution should satisfy this. Now, the question is, if one manages to find such a point, so let us say I give you a point x star , λ star and claim that it satisfies all these conditions, then can we say anything about this this point. So again, so this is an interesting result and this is slightly involved to prove so we will not really do that in this course. So in general.

So, if x star , λ star satisfies the above conditions, so again, and some regularity conditions which we will not talk about. Think of the regularity conditions as the just avoiding some kind of corner cases like so, because if you have to be like make a rigorous statement you should

have a look at the corner cases as well. But in general, the idea is that if x star λ star satisfies the above conditions.

Then we can say that the this x star is actually a local minimum. So, basically what we are saying is that so, all. So, maybe I will say that we basically if if so, this is the sufficiency part. So, this way so, this is a local minimum So, remember we said that for the unconstrained case if the gradient at x star is 0 then it is a local optima.

So, this is something similar is what we are trying to say here if x star , λ star we just call it local optima to be safe optima. So, if x star and λ star satisfies the above condition then what we have is that is a local optima. So, this is like a certificate, so, if if I can show x star λ star which has the certificate that it satisfies this condition then it necessarily, I mean then it is a local optima.

So, this is this is sufficient to say that it is a local optima. Now, put it together with the previous argument that we did where if f and g are convex, because strong duality holds these exact same conditions are necessary for optimality. Now, we are saying that it is sufficient for irrespective of it is convex or not it is sufficient for being a local minima and we know that for convex problems local minima, I mean essentially there are only global minimas.

Now, then what we are essentially saying now is that for convex problems, these conditions kind of completely characterize the optimal solution. Say, they are both necessary and sufficient. So, that is the main reason why these conditions are useful. And these conditions are what are called as the KKT conditions named after three people Karush-Khun-Tucker.

So, so, these are names of people who came up with these conditions. And these conditions kind of completely characterize the solution for convex problem. And in general, it also gives you sufficiency condition for any general problem, or any general constraint optimal problem. So, the, the KKT conditions are, I mean can also be written when you do not have just one constraint, which is the case that we have looked at so far in this discussion.

You can use the same kind of ideas go through when you have multiple inequality constraints and multiple equality constraints, just for the sake of completion, I am going to put them down. So, now, what is the general problem that we will look at is minimize over f of x, not just one inequality constraints, we will look at, let us say hi of $x \leq 0$ for several i's. So, i equals one to m, there are m such equal inequality constraints.

And let us say L J(x) is ≤ 0 for all j equals one to n, there are n inequality, equality constraints sorry, this is equal to 0, these are equality constraints. So, you have in total m + n constraints, m inequality and n equality constraints. So, this is the problem that we want to solve. And you can of course now form the Lagrangian of this.

So, which is going to be before it was just x , λ. Now, each of these in constraints will have its own Lagrange multiplier. So, it is going to be let us call them u and v. Now remember, this is a vector. These are also vectors. So, u1, u2 till um. And this is going to be v1, v2, vn. But each constraint you associate the Lagrange multiplier.

And then you write this function as f of x + sum over i equals 1 to m, ui hi of x + sum over j equals 1 to n vj L J(x). So, this is again you can now try to argue whatever we argued earlier for this new Lagrangian to go through exactly and so, the KKT conditions for this problem would look like the following. So, let me note them down separately here the first one would be the stationarity condition which would simply be λ of f of x star + sum over i equals 1 to m.

Now, remember instead of sort of tilde of f of x star the gradient + now instead of the λ we have ui's and vj's which will be the in the dual variables which will be ui star hi of x star + some over J equals 1 to n, vj star h, L J(x) star equals 0, sorry, that is the gradient, gradient. So, the gradient of the Lagrangian function at the primal optimal variable x star and the dual optimal variables u star and v star is 0.

So, this is 0 vector, this is a 0 vector. It was a vector earlier as well. So, this is condition one, condition two is our complementary slackness condition. So, that is ui star into hi of x star equals 0 for all i. Of course, this is this this has to be specified explicitly only for the inequality constraints h h hi's for the equality constraints, obviously, this is true because x star is feasible and so, L J(x) star is going to be 0.

So, you do not have to separately write this. So, this is only relevant for the inequality constraints. So, this is the complementary slackness condition CSS complementary slackness, this is the stationarity condition and we have the usual hi of x star is ≤ for our i L J(x) star equal to 0 for all j these have to be satisfied by any optimal solution these are feasible, feasibility, what feasibility these are primal feasibility and finally, the dual feasibility which just say you u i star is ≥ 0.

So, for all i these are dual feasible, feasibility. So, these are the general KKT conditions and this is this holds for as you can see any general optimization problem which have inequality constraints and equality constraints put together. So, this is the most general form of stating the KKT conditions, though we looked at the arguments with the single inequality constraints, just for the purposes of exposition.

So, you can do the same argument for the general case and it will go through. Okay good. So, now, we have kind of seen the equivalent of the first order necessary conditions and sufficiency conditions depending on the functions for constraint optimization problem and KKT conditions are the right way to look at them.

